

PGD-GP: A Chinese Named Entity Recognition Model for Constructing Food Safety Standard Knowledge Graph

Yi Chen¹, Qiuxu Fan¹, Xianpeng Yuan¹, Qinghui Zhang¹, and Yu Dong²

Abstract—The extensive range of food safety standards poses a significant challenge to efficiently accessing specific information within this domain, necessitating innovative solutions to streamline the process. In response, researchers are focusing on constructing a knowledge graph based on food safety standards to facilitate efficient associative querying. Named entity recognition is a pivotal element in this endeavor due to its critical impact on the accuracy and quality of the knowledge graph. To address the nuanced challenges of accurately identifying nested entity boundaries and rectifying entity class imbalances in food safety standards, we present PGD-GP, a novel Chinese named entity recognition model. This model is based on Projected Gradient Descent for adversarial training and Global Pointer. The model innovatively refines the Chinese Bert model at the encoding layer, employing the adversarial training method PGD to iteratively introduce perturbations to character vectors, thereby significantly enhancing the model's robustness and adaptability to texts. The decoding layer leverages Global Pointer to accurately determine dependencies and relative positional relationships between characters, thus facilitating more precise recognition of entity boundaries. To combat the issue of class imbalance, Circle Loss is utilized as the loss function. We developed and annotated the Food Safety Standard Dataset using a specifically tailored ontology rule for food safety standards. Comparative experiments conducted on the Food Safety Standard Dataset and the public Resume dataset demonstrate that PGD-GP surpasses six mainstream baseline models in performance, thereby validating the effectiveness and robustness of PGD-GP. Building upon the foundation of PGD-GP and the Food Safety Standard Dataset, we implemented a prototype system that integrates a food safety standard-based knowledge graph with associated queries. This system serves as an efficient, accurate, and comprehensive intelligent assistant, enabling researchers to effectively acquire food safety standard information.

Index Terms—Named entity recognition, projected gradient descent, global pointer, knowledge graph, food safety standards, ontology rule.

I. INTRODUCTION

THE global focus on food safety has intensified in recent years, driven by growing concerns over foodborne illnesses, environmental factors, and international trade regulations [1]. This increased awareness has led to the development and implementation of stringent food safety standards (FSS) worldwide. These standards are crucial for ensuring food quality and safety, guiding food production, and overseeing food supervision [2]. In China, the emphasis on FSS is particularly pronounced, with the number of national food safety standards reaching 1563 by September 2023 (according to the statistics of the Food Safety Standards and Monitoring and Evaluation Department, National Health Commission of P.R.China.¹). Moreover, the contents of FSS are interrelated, for example, some indicators are specified by referencing other standards, rather than presenting information explicitly. This makes it time-consuming and complicated to obtain complete and relevant information. Knowledge graph [3] can associate various types of relevant information in FSS to form a knowledge network, thus achieving precise associative query of food safety standard information [4], [5], [6], [7], helping domain researchers to obtain food safety standard knowledge efficiently [8], [9]. Therefore, it is necessary to construct a knowledge graph for FSS.

Named entity recognition (NER) aims to identify entities that belong to predefined semantic types [10] and is a key step in building a knowledge graph. The accuracy of NER directly affects the quality of downstream tasks, such as relation extraction and knowledge graph construction [11]. Therefore, entity extraction is a fundamental and critical step in knowledge extraction. NER for FSS aims to identify entities such as standard numbers, standard names, food names, indicator names, and indicator values within FSS texts. FSS texts typically contain nested entities; for example, in the standard “Cold storage for apple” [12], the phrase “Cold storage for apple” is the standard name, while “apple” is also a food name. These entities have complex nested relationships, complicating the accurate identification of entity boundaries. Additionally, there is an issue of entity class imbalance in FSS texts, where the number of entities for food names

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¹[Online]. Available: <http://www.nhc.gov.cn/sps/>

is significantly higher than that for food categories. This imbalance poses a challenge to the robustness of the entity recognition model. Current research predominantly uses rule-based methods for entity recognition [13], which have a high miss rate for nested entities in standard content and limited applicability to a broader range of food safety standards.

To address these challenges, we propose a Chinese named entity recognition model called PGD-GP, which is based on the adversarial training method PGD and Global Pointer. We annotated the corpus using a domain-specific ontology rule that we developed, leading to the creation of a dataset focused on food safety standards. By applying NER and relationship extraction to this dataset, we extracted entities and relationships, organized them into triples, and stored them in a graph database. The graph database further combines with front-end technologies to implement a knowledge graph of food safety standards in a prototype system. This provided knowledge graph facilitates associative querying by researchers for food safety standard information.

The three contributions of this work are outlined as follows:

- We presented PGD-GP, a named entity recognition model tailored to address the challenges of entity nesting and class imbalance within Chinese food safety standards. The robustness and effectiveness of the PGD-GP model were evaluated on two distinct datasets.
- We developed an ontology rule for food safety standards and established the Food Safety Standards Dataset (FSSD). This dataset serves as fundamental resource that enables researchers to construct knowledge graphs and retrieve relevant information from food safety standards.
- Based on the PGD-GP model and the FSSD, we developed a prototype system that incorporates a queryable knowledge graph focused on Chinese food safety standards. This system facilitates associative querying, thereby greatly assisting researchers in efficiently, accurately, and comprehensively accessing information related to FSS.

The rest of the sections are organized as follows: Section II introduces the related work. Section III presents the proposed PGD-GP model, including its components and working mechanism. Section IV outlines the ontology rule of the food safety standards and the dataset; Section V introduces experimental evaluation for the PGD-GP model. Section VI introduces the construction and application of the food safety standards knowledge graph. Finally, Section VII summarizes and discusses our work.

II. RELATED WORK

Although many studies are devoted to food safety research, there are few studies on information extraction and knowledge graph construction in food safety domains. This section elaborates on named entity recognition models, knowledge graph construction, and knowledge graph on food safety standards.

A. Named Entity Recognition Models

NER can be categorized into rule-based, machine learning, and deep learning methods. Initially, NER employed rule-based methods, which depended on manually crafted rules and

dictionaries for entity identification [14]. However, these rules and dictionaries were challenging to apply across different domains and languages, and they struggled to identify complex entities. To address this issue, researchers utilized statistical models and machine learning algorithms to learn entity recognition models from annotated data [15], [16], [17]. Deep learning methods have since become predominant, bypassing the complex feature engineering required in machine learning methods and resolving long-distance dependency problems [18]. These methods primarily include sequence labeling and span-based approaches.

Sequence labeling methods utilize the context information to assign a category to each token of the entity. LR-CNN [19] employed a dictionary reconstruction mechanism based on convolutional neural networks, dynamically adjusting the weights and representations of dictionary words to suit different contexts. LEBERT [20] enhanced the Bert (Bidirectional Encoder Representations from Transformers) model's representation [21] using external dictionary-based prior knowledge. Lattice LSTM [22] combined word-character mixed representations, accounting for both character-level and word-level information, which improved the model's recognition and generalization capabilities. FLAT [23] utilized the Transformer model to encode word-character information, optimizing nested entity recognition. TENER [24] implemented a hierarchical attention mechanism to capture features at different levels, effectively reducing computational complexity and enhancing multilingual and cross-domain adaptability. However, sequence labeling methods have limitations in handling nested entities and might produce inconsistent annotations for entities with ambiguous boundaries. Span classification methods emerged as a new paradigm for NER, addressing these challenges [11]. Global Pointer [25] predicts entities by considering the global view of the span's start and end positions and introduced a classification loss function to address the issue of class imbalance.

Adversarial training is a common method that trains the network using both original and adversarial samples, enhancing the robustness and generalizability of the NER model. Common adversarial training algorithms in NLP include FGSM, PGD, and FreeAT. Li et al. [26] utilized an NER model based on the Global Pointer and adversarial training to improve the model's robustness and generalization ability. However, the FGSM adversarial training algorithm they employed, while computationally efficient, lacks exploration of diversity and rationality in adversarial samples due to its single-step generation approach. This limitation hampers the improvement of the model's resistance to attacks. On the other hand, the first-order perturbation, multi-step algorithm PGD [27], employs a small-step, multi-walk strategy for adversarial training. This ensures the perturbation range remains manageable, optimizes the calculation of adversarial perturbation, and generally achieves better results.

B. Knowledge Graph Construction

A knowledge graph is a semantic network that reveals the relationships between entities [3] and has been widely applied in

fields such as sentiment detection [28], question-answering systems [29], personalized recommendation [30], [31], and others. Constructing a knowledge graph requires knowledge extraction, which encompasses two main aspects: named entity recognition and relation extraction. The objective of relation extraction is to address the semantic linkage between entities. The predominant methods in this area are based on deep learning and can be categorized into pipeline methods and joint extraction methods. Pipeline methods break down the task into two independent steps, typically employing CNNs, RNNs, and their variants for relation extraction. In contrast, joint extraction methods handle both tasks simultaneously, taking into account their mutual constraints [32], and outperform pipeline methods in addressing the issue of error accumulation. However, PURE (Princeton University Relation Extraction) [33] demonstrated that the pipeline method, which incorporates entity information into the relation model, can outperform the joint extraction method.

Knowledge graphs can be classified, based on their application scope, into general knowledge graphs and domain-specific knowledge graphs. General knowledge graphs, like DBpedia [34], offer broad coverage and large-scale datasets with a high degree of automation. Domain-specific knowledge graphs, on the other hand, focus on data from specific industries. Although smaller in scale, they provide high-quality, precise knowledge. Knowledge graphs have been constructed in various fields, including medicine [35], construction safety [36], emergency response standards [37], food [38], news [39], and more, to facilitate effective representation and application of knowledge.

C. Knowledge Graph on Food Safety Standards

The knowledge graph on food safety standards is a knowledge graph application that interlinks various information within food safety standards to form a relational network. This technology enables efficient food safety standard information services and assists food supervision institutions and researchers in effectively acquiring food safety standard knowledge. There have been some studies conducted on the construction and application of the knowledge graph on food safety standards. For instance, Lv et al. [40] proposed an open relation extraction model based on dependency parsing for the agricultural product domain and constructed a knowledge graph for the agricultural product standard domain. However, their work was solely focused on agricultural product standards and did not encompass a complete food safety standard system. Additionally, Qin et al. [13] constructed a knowledge graph of the citation relationships between the content and documents of the national food safety standards and developed a visual analysis system [41]. Nonetheless, these works primarily utilized rule-based and syntactic analysis methods, which had a high omission rate for nested entities in standard content, and the rule templates could not be easily extended to a broader range of standards in food safety domain.

Given these limitations, it is imperative to design effective knowledge extraction methods for food safety standards and construct corresponding knowledge graphs in the domain of food safety. This stands as a crucial research direction. Such efforts can not only improve knowledge management efficiency

in the food safety domain but also advance the development and application of related technologies.

III. THE NER MODEL PGD-GP

To achieve named entity recognition of texts, we presented a Chinese NER model PGD-GP based on Projected Gradient Descent and Global Pointer.

A. Principle of PGD-GP Model

The PGD-GP model consists of three parts: the pre-trained language model Bert, the adversarial training method PGD, and the decoder Global Pointer, as shown in Fig. 1. The model first uses the semantic syntactic knowledge learned by the fine-tuned Bert pre-trained model, encodes the input character sequence S , and obtains the vector representation X of each character. Since the Chinese corpus of Bert is small, the generated vectors have poor adaptability to the specific domain text. We introduce the adversarial training method PGD and iteratively add perturbations to the vector X , to obtain the adversarial sample vector H , which improves the robustness of the model to adversarial samples and allows the model to better capture and express the semantic information of character sequence. Then, to address the problem of entity class imbalance in the text, we use the Global Pointer to learn the dependency and relative position information between characters, and perform the head prediction and tail prediction modules on the vector H to obtain the vectors Q and K , respectively. We use relative position encoding to add relative position information to the vectors Q and K , and obtain the span type score by the inner product of the two vectors. We use the Global Pointer to decode the span type score into multiple span matrices. The input of the model is a character sequence, and the output is a span matrix sequence, indicating the entity type of each span.

B. Fine-Tuning Bert

In common NER tasks, position and context semantic information play a key role. General language models such as N-gram, Word2vec, and other static language models cannot well understand the contextual relationship between sentences and cannot represent polysemous words. Bert is a Bidirectional Encoder based on Transformer, which has the ability to capture contextual information and provide positional information. Therefore, we choose Bert as the pre-trained model and use it to dynamically generate word vectors with contextual information.

First, we fine-tune the Bert model on the target text dataset to obtain the vector representation of the text. During the fine-tuning, the Bert model learns how to identify various named entities from the input text data. The input vector of Bert consists of three parts: Token Embeddings, Position Embeddings, and Segment Embeddings. Given a sentence $S = [s_1, s_2, \dots, s_{(n-2)}]$ with $n - 2$ tokens, we first add special markers $[CLS]$ and $[SEP]$ at the beginning and end of the sentence, then convert each token into its corresponding word vector, and add position vectors and segment vectors to obtain the input representation $E = [e_1, e_2, \dots, e_n]$. The input is fed into the Bert

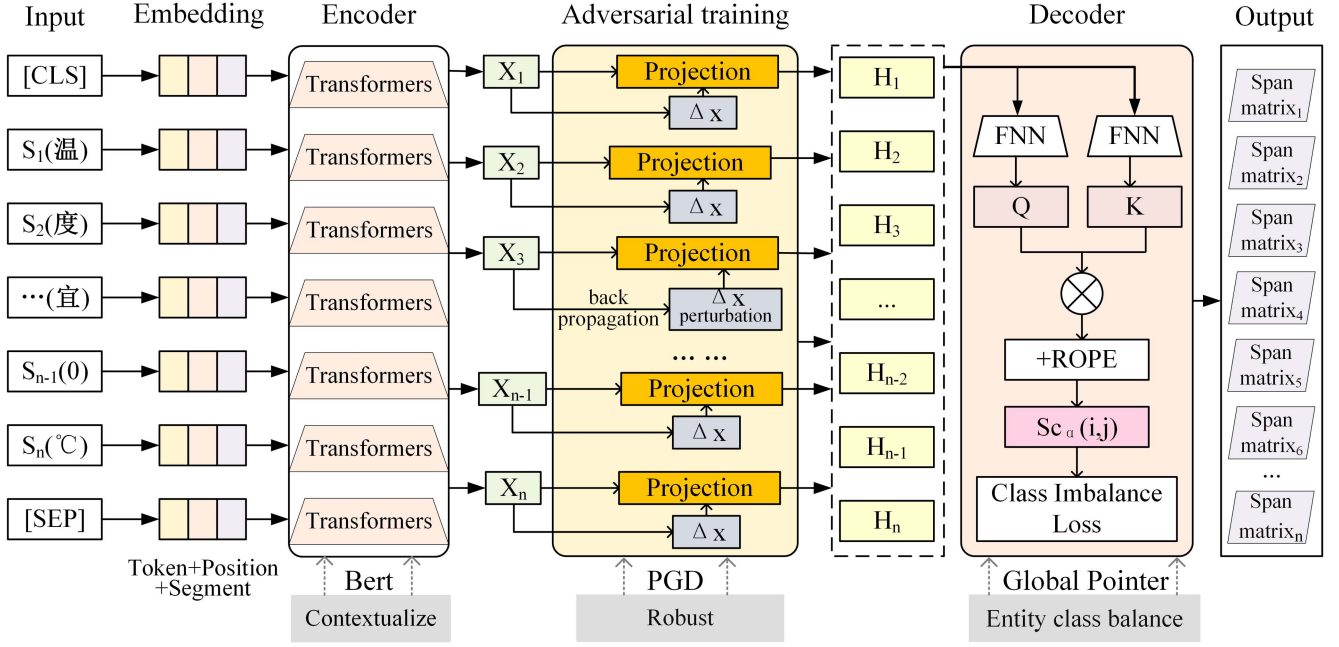


Fig. 1. Structure illustration of the PGD-GP model for named entity recognition in Chinese text. The input S_1-S_n represents an example sentence: ‘The temperature should be 0 °C’ from a certain food safety standard.

module, which outputs the context-aware representation of each token $X = [x_1, x_2, \dots, x_n]$, where $X \in \mathbb{R}^{n \times v}$, v is the vector dimension, and x_i is the vector representation of the i th token, $i = 1, \dots, n$.

C. PGD in Adversarial Training

Due to the small size of the Chinese corpus for Bert, the word vectors generated by it are sensitive to the noise around the entities and have poor adaptability to the FSS domain text. To improve the robustness of the model and prevent it from producing wrong classification results when facing noise, we use PGD to add perturbed samples during the model training process, so that the model can better capture and express the semantic information of the FSS character sequence. PGD (Projected Gradient Descent) is a gradient-based adversarial sample generation method, which is based on the idea of perturbing the input sample X along the gradient direction of the loss function $L(\theta, x, y)$ at each iteration, and projecting the perturbed sample into an allowed perturbation range R , thus obtaining a more adversarial sample H .

Specifically, we first add a random perturbation r within the neighborhood range R of the original sample X as the initial adversarial sample X_0^{adv} , where $r \in R$, as shown in (1):

$$X_0^{adv} = X + r \quad (1)$$

Then we iteratively use X_t^{adv} to perform adversarial attacks, calculate the loss $L(\theta, x, y)$ after adding perturbation, backpropagate to calculate the current loss function with respect to the input gradient $\nabla X_t^{adv} L(\theta, X_t^{adv}, y)$, and calculate the new perturbation ΔX by (2).

$$\Delta X = \varepsilon \cdot \text{sign} [\nabla X_t^{adv} L(\theta, X_t^{adv}, y)] \quad (2)$$

Where ε is the perturbation intensity, sign is the sign function, and $\nabla X_t^{adv} L(\theta, X_t^{adv}, y)$ is the gradient of the loss function relative to the current input X_t^{adv} . We add the perturbation ΔX to X_t^{adv} , and project the updated adversarial sample onto a range R near X to ensure that it does not deviate too far from the original distribution. Then the adversarial sample generated by iterating $t + 1$ times is as shown in (3):

$$X_{(t+1)}^{adv} = \Pi_{(X+R)} \{X_t^{adv} + \varepsilon \cdot \text{sign} [\nabla X_t^{adv} L(\theta, X_t^{adv}, y)]\} \quad (3)$$

Where Π is the projection function. Repeat the above steps until the preset number of iterations or perturbation range is reached, and obtain the adversarial sample vector $H = X^{adv}$ as the training sample input to the decoder for training. In this way, we can optimize the model on the adversarial samples and improve the model’s robustness and generalization ability on the original samples.

D. Enhancing NER Performance With Global Pointer

The Global Pointer (GP) decoder is employed to identify the boundaries and types of entities, thus improving the accuracy of entity recognition and effectively addressing the problem of entity class imbalance in datasets. GP operates as a span-based decoding method capable of extracting the boundaries and types of named entities from the vectors output by Bert. Its principal strategy is to utilize relative position information to improve the representation and classification of spans. This is achieved by encoding the relative position information through a multiplication attention mechanism and by using a general loss function to resolve the class imbalance issue. GP considers the start and end positions of entities from a global perspective, treating the head

and tail as a unified entity, which ensures consistency between the training and inference processes.

The basic idea of GP is to identify entity spans as basic units. For a sentence of length n , it constructs C span matrices of size $n \times n$ (upper triangular matrices containing valid entity information) to traverse all valid spans, corresponding to C types of entities in the sentence. Each element of the matrix corresponds to a possible span, where C is the number of entity categories. The row index of the cell is the head position of the entity, and the column index is the tail position.

Given an input sequence S of length n , which is encoded by the Bert and PGD modules to obtain a vector sequence $H = [h_1, h_2, \dots, h_n]$, the GP model first uses two feedforward networks to calculate the vector representations $q_{(i,\alpha)}$ and $k_{(i,\alpha)}$ of each word as the head and tail of a span, as shown in (4) and (5).

$$q_{(i,\alpha)} = W_{(q,\alpha)}h_i + b_{(q,\alpha)} \quad (4)$$

$$k_{(i,\alpha)} = W_{(k,\alpha)}h_i + b_{(k,\alpha)} \quad (5)$$

Where $q_{(i,\alpha)} \in \mathbb{R}^d$, $k_{(i,\alpha)} \in \mathbb{R}^d$ are the vector representations of the i th word as the head or tail of an entity of type α , $W_{(q,\alpha)}, W_{(k,\alpha)} \in \mathbb{R}^{(v \times d)}$ and $b_{(q,\alpha)}, b_{(k,\alpha)} \in \mathbb{R}^d$ are learnable parameters, v is the dimension of the Bert output vector, and d is the dimension of the head or tail vector. Thus, we obtain the vector sequences $Q_\alpha = [q_{(1,\alpha)}, q_{(2,\alpha)}, \dots, q_{(n,\alpha)}]$ and $K_\alpha = [k_{(1,\alpha)}, k_{(2,\alpha)}, \dots, k_{(n,\alpha)}]$ for identifying entities of type α , and then we calculate the score of each span $t[i : j]$ as an entity of type α according to (6):

$$\text{Sc}_\alpha(i, j) = q_{(i,\alpha)}^\top k_{(j,\alpha)} \quad (6)$$

We use the inner product of $q_{(i,\alpha)}$ and $k_{(j,\alpha)}$ as the score of span $t[i : j]$ being an entity of type α , where $t[i : j]$ refers to the continuous span composed of the i th to j th elements of sequence t .

To use relative position information, we explicitly inject relative position information into the model, use ROPE (Rotary Position Embedding) [42] to encode the position of each word in the sentence, and multiply it with the head or tail vector. ROPE uses absolute position encoding to achieve relative position encoding. ROPE satisfies $R_i^\top R_j = R_{(j-i)}$, that is, relative position encoding only depends on the span length. We apply it to the entity representations Q_α and K_α . Thus, our scoring function is shown in Equation (7).

$$\text{Sc}'_\alpha(i, j) = (R_i q_{(i,\alpha)})^\top (R_j k_{(j,\alpha)}) = q_{(i,\alpha)}^\top R_{(j-i)} k_{(j,\alpha)} \quad (7)$$

GP treats entity recognition as a multi-label classification problem, and uses a unified loss function proposed in Circle Loss [43] to suit the situation where the total number of categories is large and the number of target categories is small. It takes all possible entity spans in each sentence as a candidate set, and then calculates the score of each span for each entity type separately, which reflects the probability of the span belonging to that type. To solve the class imbalance problem, it uses a positive sample set P_α that contains the head and tail positions of all entities of type α , and a negative sample set N_α that contains the head and tail positions of all non-entities or entities of non- α

type. Then it minimizes the loss function, as shown in (8).

$$\log \left(1 + \sum_{(i,j) \in P_\alpha} e^{-\text{Sc}'_\alpha(i,j)} \right) + \log \left(1 + \sum_{(i,j) \in N_\alpha} e^{\text{Sc}'_\alpha(i,j)} \right) \quad (8)$$

Where $\text{Sc}'_\alpha(i, j)$ represents the score of the span composed of the i th to j th elements in the input sequence belonging to type α . During decoding, a simple threshold of 0 is used to determine which spans are target categories, that is, output all spans that satisfy $\text{Sc}'_\alpha(i, j) > 0$ as entities of type α . This method can effectively use global information while avoiding complex decoding processes.

IV. FOOD SAFETY STANDARD ONTOLOGY RULE AND DATASET

To evaluate our proposed PGD-GP model and further facilitate the construction of an FSS-KG for researchers, we detail the construction process of the food safety standard ontology and dataset in this section.

A. Ontology Rule Construction of FSS

The construction of the ontology rule is a pivotal task in developing the FSS-KG. Considering the wide range of FSS types, the structure of both content and basic information tends to be similar across various documents, with differences mainly in specific regulations or standards. Consequently, we chose representative documents from food cold chain standards as examples for our ontology rule research and partially translated them into English to facilitate better comprehension by readers. The key points of information that researchers focus on include:

1) *Standard Content Information*: This encompasses the food name, food category, indicator name, and indicator value mentioned in the standard. These pieces of information reflect the quality requirements and safety assurances for food, such as temperature, humidity, storage period, and other significant indicators. We break down the indicators into two basic entities: indicator name and indicator value. By identifying them as separate entities, each category possesses distinct features, enhancing the recognition accuracy for each category and, consequently, the overall indicator recognition accuracy. For instance, in the national standard “GB/T 8559-2008 (Cold storage for apples),” the food name is “apple”, and the sentence “The storage temperature of apples should be determined according to different varieties, with most being suitable for storage between -1 °C and 0 °C” specifies “storage temperature” as the indicator name and “-1 °C to 0 °C” as the indicator value.

2) *Standard Basic Information*: This includes the standard number, standard document name, main drafters, drafting department, responsible department, and other relevant details. These pieces of information reflect and distinguish the source and authority of each food safety standard and are essential for retrieving and citing each standard.

The entity design for food safety standards was constructed through analysis, as detailed in Table I. We clearly designed eight types of entities as: indicator name, indicator value, food name,

TABLE I
ENTITY DESIGN IN FSS

Entity Label	Entity Name	Entity Meaning	Entity Examples
IN	Indicator Name	Names of indicators for food	“Temperature”, “Humidity”, “Storage Quantity”
IV	Indicator Value	Values or ranges of indicators (with units)	“-1°C to 0°C”, “80% to 90%”, “8% to 5% of storage capacity”
FN	Food Name	Names of food items mentioned in the standard	“Apple”, “Cucumber”, “Lettuce”,
FC	Food Category	Categories of food items covered by the standard	“Fruits and Vegetables”, “Fruit”, “Perishable Foods”
SNum	Standard Number	Number assigned to a standard document	“GB/T 8559-2008”, “WB/T 1059-2016”
SN	Standard Name	Title of the standard document	“Apple refrigeration technology”
ORG	Organization	Organization responsible for drafting or authorizing the standar	“Chinese National Federation of Cooperative Supply and Marketing”
PER	Person	Individuals involved in drafting the standard	“Ye Qiteng”, “Qin Yuming”

TABLE II
RELATIONSHIP DEFINITION IN FSS

Relationship	Relationship Meaning	Subject	Object	Relationship Example
Contain	Indicators contained within the food/food category/indicator name	FN/FC/IN	IN	(Apple → Storage Temperature)
Value	Values of the indicator name	IN	IV	(Storage Temperature → 0°C to 1°C)
Correspond	Standard number corresponding to the name of the standard	SNum	SN	(GB/T 8559-2008 → Cold storage for apple)
Regulate	Foods or food categories specified in the standard	SNum	FN/FC	(GB/T 8559-2008 → Apple)
Cite	Another standard is cited in the standard	SNum	SNum	(GB/T 8559-2008 → GB/T 10651)
Replace	The standard replaces the old one	SNum	SNum	(GB/T 8559-2008 → GB/T 8559-1987)
Authorize	Organization responsible for managing this standard	SNum	ORG	(GB/T 8559-2008 → Jinan Fruit Research Institute, Chinese National Federation of Cooperative Supply and Marketing)
Propose	Organization proposing the standard	SNum	ORG	(GB/T 8559-2008 → Chinese National Federation of Cooperative Supply and Marketing)
Draft	Organization or person who drafting the standard	SNum	ORG/PER	(GB/T 8559-2008 → Ye Qiteng)

food category, standard number, standard name, organization, and person.

To describe the semantic associations between different entities involved in FSS, we designed nine types of relations in FSS based on the semantic and syntactic features in FSS documents, which are: Contain, Value, Correspond, Regulate, Cite, Replace, Authorize, Propose, and Draft. Table II shows the meanings and examples of each relation. For example, there is an inclusion relation between “apple” and “storage temperature,” indicating that apple cold storage technology includes storage temperature as an indicator; there is a value relation between “storage temperature” and “-1 °C to 0 °C,” indicating that the storage temperature of apples should be controlled within this range. In addition, since some indicators may have other indicators below them, such as “error” having “temperature error” and “humidity error,” we also added an inclusion relation between two indicators. In the national standard “GB/T8559-2008 Apple Cold Storage Technology,” the standard number has a proposal relation with “Chinese National Federation of Cooperative Supply and Marketing,” indicating that the standard was proposed by this organization; the standard number has a regulation relation with “Apple,” indicating that the standard applies to apple cold storage technology.

B. Food Safety Standard Dataset and Annotation Method

We retrieved the documents that cover national, industrial and local standards from websites such as Food Partner Network,²

²[Online]. Available: <http://foodmate.net/>

and converted the downloaded standard PDF files into a corpus of about 220,000 characters. Corpus annotation is an important step for NER, and different characteristics of corpus require different annotation methods. The traditional sequence annotation method is not suitable for multiple nested entities in FSS corpora, and this problem can be solved by using pointer labeling methods. In the pointer labeling system, the decoder uses a span module instead of a sequence decoding module, which speeds up the training process. This method labels the entity position by setting the start position pointer and the end position pointer, and also labels the entity type at the same time. For example, “storage temperature” and “0 °C–1 °C” are labeled as two entities of indicator name (IN) and indicator value (IV), respectively. Based on the ontology rules in food safety standards, the data in the corpus are annotated with pointers, resulting in the FSSD.

V. EXPERIMENTS AND EVALUATION

After finalizing the dataset construction of Chinese food safety standards, we utilized FSSD, along with an additional Resume dataset, to evaluate our proposed PGD-GP model through comparative and ablation experiments.

A. Experimental Datasets

To evaluate our proposed PGD-GP model and its robustness and effectiveness, we prepared two distinct datasets. The first is the FSSD, which was constructed using the methods outlined in Section IV. The second dataset is the Resume dataset [22],

TABLE III
NUMBER OF EIGHT TYPES OF ENTITIES IN FSSD

Dataset	Total entities	IN	IV	FN	FC	SNum	SN	ORG	PER
Training set	7795	1529	951	1847	878	1367	396	380	447
Validation set	969	191	118	230	109	169	49	47	56
Test set	969	146	70	274	120	162	39	38	78

TABLE IV
NUMBER OF EIGHT TYPES OF ENTITIES IN RESUME

Dataset	Total entities	Name	Country	Education	Location	Organization	Race	Professional	Job title
Training set	13438	952	260	858	42	4611	115	287	6308
Validation set	1609	110	33	218	2	523	15	18	690
Test set	1736	112	28	218	6	553	14	33	772

a public Chinese-language resource that similarly presents the problem of entity class imbalance.

1) *FSSD*: The FSSD is divided into training, validation, and test sets by the ratio of 8:1:1, and the data statistics are shown in Table III.

2) *Resume Dataset*: The Resume dataset comprises 1027 resume summaries, annotated with entities in eight categories: name, country, education, location, organization, race, professional, and job title. Similarly, the Resume dataset is partitioned into training, validation, and test sets in an 8:1:1 ratio, with specific data statistics presented in Table IV.

As shown in Tables III and IV, both datasets exhibit the problem of entity class imbalance. Specifically, in the FSSD, categories such as food name and indicator name have a significantly larger number of entities compared to categories like food category and indicator value. Similarly, in the Resume dataset, categories such as job title and organization contain notably more entities than those such as country, education, and race. This imbalance of entity classes presents a challenge to the robustness of our model.

B. Experiments Configuration

All experiments were conducted in a consistent experimental environment to ensure result comparability. We implemented these experiments using the PyTorch framework and carried out the training and testing on an NVIDIA GeForce RTX 2080 Ti. The AdamW optimizer was utilized, with a learning rate of $5e-5$, a batch size of 16, a maximum sentence length of 200, and a total of 50 epochs. For our model, we employed the pre-trained Bert-base-Chinese language model, which features a hidden size of 768, 12 hidden layers, and 12 attention heads. Additionally, we applied the PGD adversarial method for adversarial training, setting the adversarial perturbation intensity to 0.3 and the number of adversarial iterations to 3.

C. Comparison Experiment

To verify the robustness and effectiveness of the PGD-GP model, we selected the mainstreaming methods as baseline models for comparison: the classic LR-CNN [19] and TENER [24] methods, the widely-used Chinese NER methods LEBERT [20]

TABLE V
EXPERIMENTAL RESULTS OF PGD-GP MODEL AND BASELINE MODELS ON FSSD AND RESUME DATASETS

Model	FSSD			Resume		
	Prec.	Rec.	F1	Prec.	Rec.	F1
Lattice LSTM	88.45	86.22	87.30	94.81	94.16	94.46
LR-CNN	89.23	87.42	87.93	95.37	94.84	95.11
FLAT	92.45	90.66	91.53	95.93	95.62	95.33
LEBERT	90.84	89.04	89.94	96.24	96.11	95.97
TENER	93.72	92.14	93.43	95.61	95.47	95.32
AT-CBGP	94.06	93.47	93.89	96.38	96.63	96.48
PGD-GP(ours)	95.40	94.53	94.52	96.54	96.41	96.58

Note: The bold values represent the best results under specific indicators in all comparison methods.

and Lattice LSTM [22], the nested entity-oriented method FLAT [23], and the latest method AT-CBGP [26].

We adopted precision, recall, and F1 score as evaluation metrics and calculated the average values for all entity types on each metric. A strict matching criterion was applied for these metrics, requiring that the predicted entity exactly match the true entity (both in type and boundary) to be considered correct. Table V presents the experimental results of our PGD-GP model and the baseline models on the FSSD and Resume datasets.

As shown in Table V, the proposed PGD-GP model surpasses the baseline models in almost all metrics on both datasets. Utilizing GP as a decoder enables us to consider the start and end positions of entities from a global perspective, treating them as a unified entity for discrimination. This approach ensures consistency in both the training and inference processes. Additionally, the integration of an adversarial training module addresses the issue of local instability in the model.

On the FSSD dataset, while AT-CBGP achieved the highest F1 score, our model surpassed AT-CBGP by 0.63% in terms of F1 score. AT-CBGP employs a one-step adversarial perturbation, which may not always reach the optimal solution within its constraints. In contrast, our model introduces the PGD adversarial method, utilizing a small-step, multi-step strategy for the adversarial attack. This ensures a moderate perturbation range, optimizes the calculation of adversarial perturbation, and enables more accurate identification of entities in the FSS

TABLE VI
THE F1 SCORE OF THE PGD-GP MODEL AND BASELINE MODELS FOR DIFFERENT ENTITY TYPES IN FSSD

Model	IN	IV	FN	FC	SNum	SN	ORG	PER
Lattice LSTM	88.53	83.75	83.35	63.77	96.33	93.98	95.83	92.88
LR-CNN	90.43	83.98	83.26	63.69	97.19	93.98	96.18	94.79
FLAT	92.86	88.97	85.67	73.08	98.85	97.56	97.36	97.93
LEBERT	91.73	85.79	84.63	68.73	97.51	96.87	97.72	96.57
TENER	93.15	90.83	86.61	81.91	99.38	98.46	98.12	98.94
AT-CBGP	93.38	90.95	86.26	85.57	99.45	98.15	98.19	99.18
PGD-GP(ours)	93.59	91.35	86.44	88.35	99.68	98.86	98.24	99.66

Note: The bold values represent the best results under specific indicators in all comparison methods.

files, reducing the likelihood of missed or misidentified entities. Although TENER and FLAT have improved model performance through enhanced relative position encoding, our model achieves comparable effectiveness without relying on additional dictionary-based support, demonstrating its efficiency.

Applied to the Resume dataset, models such as Lattice LSTM, LR-CNN, and LEBERT integrated lexical information in various ways. LEBERT utilized a dictionary adapter to inject dictionary information between the transformer layers, thereby achieving a higher F1 score than Lattice LSTM, LR-CNN, and LEBERT. AT-CBGP employed GP as a decoder, effectively leveraging relative positional information to enhance the representation and classification of spans, which resulted in a recall of 96.63% and an F1 score of 96.48%. Our PGD-GP model, while exhibiting a slightly lower recall compared to AT-CBGP, achieves better precision and an improved F1 score. This result demonstrates that our model has strong generalization and adaptability capabilities.

Additionally, Table VI presents the F1 scores of the PGD-GP model and various baseline models for different entity types in FSSD. Notably, for the less frequent entity types such as ORG and PER, and especially for FC, the F1 score of the proposed model is significantly higher than that of most baseline models. This indicates the PGD-GP model's enhanced capability in addressing the entity class imbalance problem.

D. Ablation Experiment

To further verify the effectiveness of the proposed PGD-GP model and evaluate the contribution of each component to its overall performance, we conducted ablation experiments on both the FSSD and Resume datasets. The results of these experiments are displayed in Table VII. Furthermore, to ascertain the advantages of incorporating adversarial training in the entity recognition task, we compared the performance of the model with and without PGD across different entity categories in the FSSD dataset, as illustrated in Table VIII. This comparison reveals that the addition of PGD notably improves accuracy, particularly for the less accurately identified entity categories IN, IV, and most significantly for the less frequent entity category FC in our manual annotation.

Analyzing the results presented in Tables VII and VIII leads to several key insights:

- 1) Bert provides rich semantic information, which is particularly beneficial when FSS contains a large number of professional terms and entity information. It offers high-quality text vector representations for the subsequent decoder. In terms of model initialization and performance

TABLE VII
ABLATION EXPERIMENT RESULTS

Model	FSSD			Resume		
	Prec.	Rec.	F1	Prec.	Rec.	F1
ERNIE	94.68	92.23	93.44	94.48	91.28	95.36
Random	91.05	89.28	90.16	91.73	88.3	91.37
- PGD	92.38	89.69	90.97	94.98	96.56	95.93
SpanBERT	94.22	91.73	92.96	94.81	93.27	95.43
CRF	91.18	88.72	89.94	93.16	90.92	92.89
+CE loss	95.04	92.86	94.21	95.56	92.5	95.01
PGD-GP(ours)	96.4	94.42	94.52	96.54	96.41	96.58

Note: "-" indicates the removal of the item that follows it. "+" indicates the addition of the item following it. ERNIE and Random refer to replacing the Bert encoder with ERNIE and random embeddings, respectively. SpanBERT and CRF represent replacing the GP decoder with a span-based classifier and a sequence tagger, respectively. CE loss denotes the use of cross-entropy loss instead of imbalanced loss.

The bold values represent the best results under specific indicators in all comparison methods.

TABLE VIII
THE EXPERIMENTAL RESULTS OF THE PGD-GP MODEL WITH AND WITHOUT PGD ON VARIOUS ENTITY TYPES IN FSSD

Entity Label	PGD-GP			-PGD		
	Prec.	Rec.	F1	Prec.	Rec.	F1
IN	96.41	90.93	93.59	88.52	85.71	87.1
IV	92.94	89.81	91.35	86.21	80.65	83.33
FN	89.94	86.77	86.44	88.43	84.89	86.62
FC	90.33	86.45	88.35	73.91	63.3	68.19
SNum	99.69	99.68	99.68	99.68	99.68	99.68
SN	100	98.74	98.86	96.63	97.73	97.18
ORG	99.02	97.47	98.24	98.78	95.29	97.01
PER	99.33	100	99.66	99.25	100	99.63

Note: "-" indicates the removal of the item that follows it.

- enhancement, Bert demonstrates clear advantages over ERNIE and randomly initialized word vectors (Random).
- 2) PGD adversarial training enhances the robustness and generalization ability of the model, enabling it to resist noise and improve classification accuracy. Experimental results indicate that models using PGD as adversarial training outperform in all evaluation metrics. Practically, reliance solely on original data for training can lead to overfitting or underfitting due to irregularities, incompleteness, and inconsistencies in the input text.
- 3) GP effectively utilizes relative position information, significantly enhancing entity boundary recognition. It effectively addresses issues of multiple nested entities and entity class imbalances in FSS. Compared to SpanBERT and traditional decoders, GP better captures long-distance dependencies and complex text features, mitigating problems like gradient vanishing or explosion.

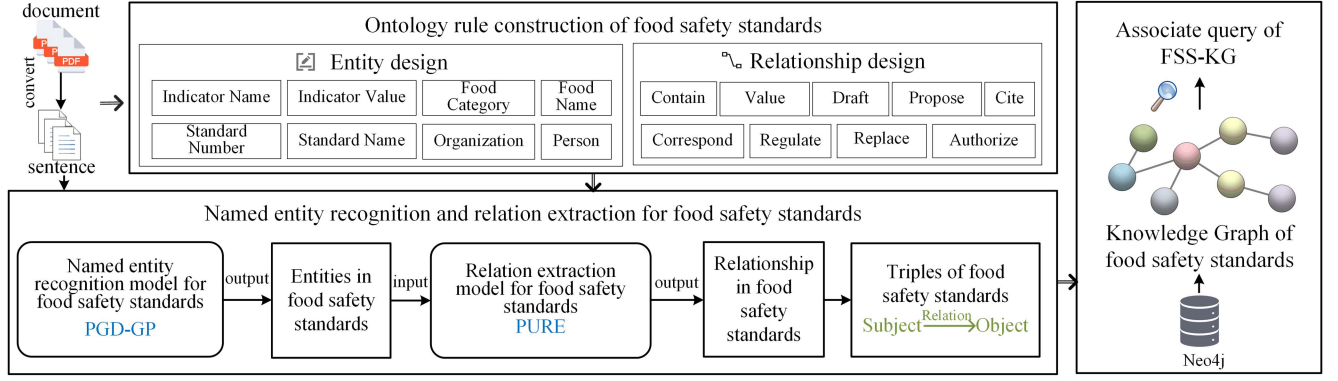


Fig. 2. Construction and application pipeline of food safety standard knowledge graph based on PGD-GP.

- 4) Imbalanced loss improves the model's ability to distinguish between different entity categories, preventing bias towards common entity categories and overlooking rare ones. Consequently, the performance of the model using imbalanced loss is superior to that of the ablation model using cross-entropy loss.

VI. FOOD SAFETY STANDARD KNOWLEDGE GRAPH

To efficiently obtain food safety knowledge through associate queries, we constructed and applied a food safety standard knowledge graph based on PGD-GP. The overall pipeline is shown in Fig. 2. Initially, we converted the FSS documents into sentence-level texts and analyzed their contents to build an ontology rule. We defined eight types of entities and nine types of relationships between entities. The texts were annotated according to this ontology rule to obtain the FSSD. Using the PGD-GP model, we extracted the FSS entities and predicted the relations between entities using the PURE relation extraction model. After processing, we obtained the FSS triplets. These triplets were then imported into Neo4j to store the FSS-KG.

A. Relationship Extraction

To further extract the relations between the food safety standard entities extracted from the text, we adopt the relation extraction method of PURE [33]. This method adds type markers at the input layer to highlight the candidate entity pairs and their entity types, and then uses Bert encoder and feed-forward neural network to predict the relation type, thus generating relation triples. Specifically, given an input sequence $\hat{X} = \{x_1, x_2, \dots, x_n\}$ and a candidate entity pair s_i, s_j , where the entity types of s_i, s_j are $e_i, e_j \in \varepsilon$, where ε is the entity type set. Define the type marker symbols $\langle S : e_i \rangle \langle /S : e_i \rangle \langle O : e_i \rangle \langle /S : e_i \rangle$, insert them before and after s_i, s_j , to indicate that they are subject and object, and mark their entity types. This results in a new input sequence:

$$\hat{X} = \{ \dots, \langle S : e_i \rangle, x_{\text{start}(i)} \dots x_{\text{end}(i)}, \langle /S : e_i \rangle \dots \langle O : e_j \rangle, x_{\text{start}(j)} \dots x_{\text{end}(j)}, \langle /O : e_j \rangle \dots \} \quad (9)$$

where $\text{start}(i)$ and $\text{end}(i)$ are the start and end positions of s_i in X , $\text{start}(j)$ and $\text{end}(j)$ are the start and end positions of s_j in X . Next, we use the pre-trained encoder Bert to encode $\hat{X}$,

obtaining its output representation, i.e.,

$$\{\hat{X}_1, \hat{X}_2, \dots, \hat{X}_n\} \quad (10)$$

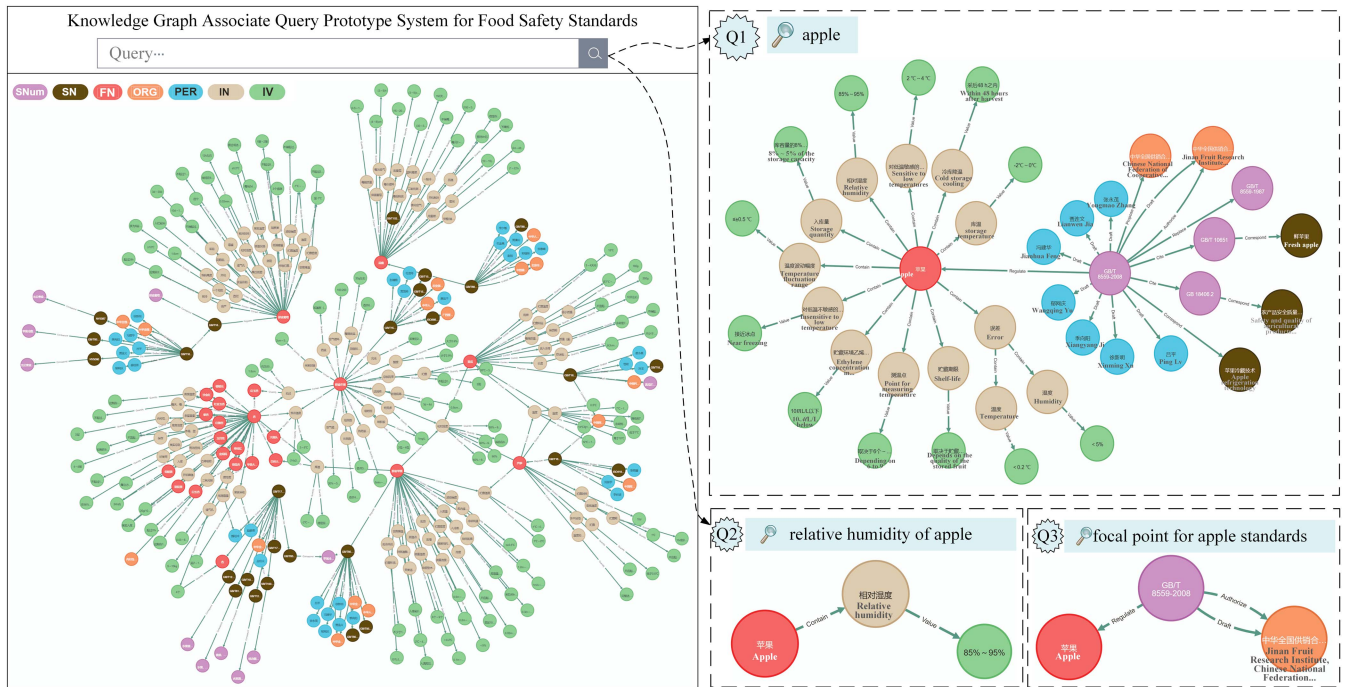
Then, we concatenated the output representations of the start positions of the subject and object as the representation of the candidate entity pair:

$$h_r(s_i, s_j) = [\hat{X}_{\text{start}(i)}, \hat{X}_{\text{start}(j)}] \quad (11)$$

where $\text{start}(i)$ and $\text{start}(j)$ are the indices of $\langle S : e_i \rangle$ and $\langle O : e_j \rangle$ in $\hat{X}$. Finally, the representation $h_r(s_i, s_j)$ is fed into a feed-forward network to predict the probability distribution of the relation type $r \in \mathcal{R} \cup \{\epsilon\}$: $P_r(r | s_i, s_j)$, where $\mathcal{R}$ is the relation type set, and ϵ is the empty relation.

B. FSS-KG Construction and Prototype System

To enable the tasks of visualizing and querying the knowledge graph in real, we integrated the triplet data derived from relation extraction. A total of 399 entities and 409 triples were imported into the Neo4j database. We then developed a prototype system by connecting the Neo4j database with D3.js to implement a front-end interface, utilizing the Python Flask framework. This prototype system visually constructs a knowledge graph for food safety standards, enabling researchers to conduct intelligent and iterative queries. As depicted in Fig. 3, the prototype system encodes circles to represent entities in the knowledge graph, with different colors signifying different entity types. The graph on the left illustrates the overall structure of FSS-KG, highlighting the extensive associative relationships within FSS. According to type-in text, it is possible to query association information about related entities. For instance, the right graph Q1 shows a query for entities and relationships related to “apple” in the graph, allowing for a comprehensive display of all indicator contents and standard information pertinent to apples. More specific queries can also be executed; for example, Q2 represents a view of the relative humidity value for apples, retrieving a specified range of 85%–95%, while Q3 displays the standard authority unit for apples, revealing that the authoritative unit is the Jinan Fruit Research Institute, Chinese National Federation of Cooperative Supply and Marketing. Querying within FSS-KG allows domain researchers or storage and transportation supervisors to access



researchers to query food safety standard information efficiently and accurately.

This paper presents an effective solution for the NER task and knowledge graph construction in the realm of food safety standards, providing essential technical support for the extraction of information from FSS files. This has significant implications for ensuring food quality and safety and promoting the development of the food industry. The model proposed here also offers a novel approach for NER in other domains, suitable for diverse and structurally complex entity recognition tasks, enhancing the quality and efficiency of entity recognition. Although this model is designed for Chinese, its approach has some generality for other languages. It can process texts in different languages using the multilingual version of Bert with minimal modifications. However, adaptations and optimizations may be required for different domains, tasks, and data. Additionally, the model faces challenges, such as leveraging cross-lingual and multi-modal knowledge to improve performance. These future works merit further exploration and resolution.

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